Chapter test

33 Regulation of water content

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following correctly arranges the blood vessels in an ascending order of urea concentration in the blood?

   A. hepatic vein, renal vein, renal artery
   B. renal vein, renal artery, hepatic vein
   C. renal artery, hepatic vein, renal vein
   D. hepatic vein, renal artery, renal vein ☐

2. In a healthy person, which of the following substances is/are normally absent in the urine?

   (1) glucose
   (2) amino acids
   (3) mineral salts

   A. (1) only
   B. (2) only
   C. (1) and (2) only
   D. (1), (2) and (3) ☐

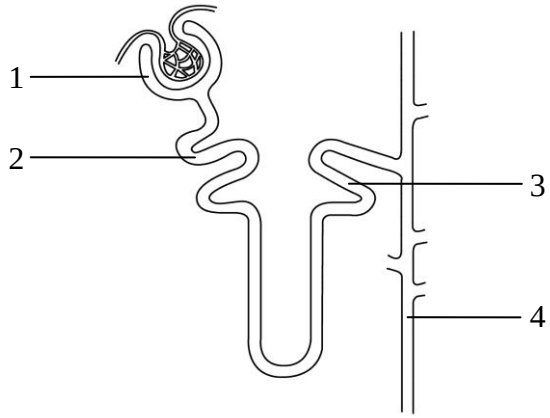
★ 3. Which of the following statements correctly compare the renal artery and the renal vein?

   (1) The blood glucose concentration in the renal artery is the same as that in the renal vein.
   (2) The blood in the renal artery has a higher oxygen content than that in the renal vein.
   (3) The blood in the renal vein has a lower urea concentration than that in the renal artery.

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3) ☐

© 2019 Aristo Educational Press Ltd.

Directions: Questions 4 and 5 refer to the diagram below, which shows the structure of a nephron.



4. Which region of cells contains the largest amount of mitochondria?

A. 1
B. 2
C. 3
D. 4 ☐

5. Which region(s) of the nephron is/are responsible for the reabsorption of water and glucose respectively?

	Reabsorption of water	***Reabsorption of glucose***
A.	1 and 3	4
B.	2	2 and 4
C.	2 and 4	2
D.	4	1 and 2 ☐

6. Which of the following comparisons between ultrafiltration and reabsorption is correct?

	Ultrafiltration	***Reabsorption***
A.	occurs in the glomerulus	occurs only in the collecting duct
B.	involves filtration under pressure	involves diffusion, osmosis and active transport
C.	results in the formation of glomerular filtrate	results in the formation of tissue fluid
D.	from nephron to blood	from blood to nephron ☐

© 2019 Aristo Educational Press Ltd.

7. Which of the following statements about the antidiuretic hormone is/are correct?

(1) ADH is produced by the pituitary gland.
(2) ADH is transported to the kidneys by blood.
(3) ADH increases the permeability of the collecting duct to water.

A. (1) only
B. (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

★ 8. Which of the following would lead to an increase in the release of antidiuretic hormone?

(1) increased water intake from drinking distilled water
(2) a fall in the water potential of the blood
(3) increased water loss due to diarrheoa

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

★ 9. Which of the following correctly states the changes in the release of antidiuretic hormone and the concentration of urine after drinking a large amount of water?

	Release of ADH	*Concentration of urine*
A.	increases	increases
B.	decreases	increases
C.	increases	decreases
D.	decreases	decreases ☐

10. Which of the following statements concerning haemodialysis in a kidney machine is correct?

A. The dialysis tubing is permeable to urea only.
B. The dialysis fluid usually flows in the opposite direction to the blood flow.
C. The incoming dialysis fluid does not contain glucose.
D. Proteins can diffuse out of the blood into the dialysis fluid. ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	33.2	33.3	33.2	33.3	33.3	33.3	33.4	33.4	33.4	33.5

© 2019 Aristo Educational Press Ltd.

B. Problem solving (21 marks)

★ 11. The table below shows the amounts of glucose, protein and urea in the fluid collected from different sites in a kidney.

Sample sites in the kidney	Amount (arbitrary unit)			Flow rate (cm^3 min^{-1})
	Glucose	Protein	Urea	
Filtrate in Bowman's capsule	800	0	54	125
Filtrate at the end of the first coiled tubule	0	0	27	25
Filtrate at the beginning of the collecting duct	0	0	27	2

(a) Based on the table, suggest the process by which the following substances are reabsorbed. Give a reason to support your answer.

(i) Glucose (2 marks)

(ii) Urea (2 marks)

(b) Explain why proteins are normally absent in the glomerular filtrate. (1 mark)

(c) Suggest an explanation for the decrease in the flow rate as the filtrate moves along the nephron. (2 marks)

© 2019 Aristo Educational Press Ltd.

(d) The above data were collected from a kidney maintained at 37 ℃. When the kidney was maintained at 30 ℃, the amount of glucose at the end of the first coiled tubule was 40 arbitrary units. Suggest an explanation for this result. (3 marks)

(Total: 10 marks)

★ 12. Twenty healthy adults participated in an investigation. They were divided into two groups (A and B). At the beginning, their urine samples were collected and analysed. After 60 minutes, the participants in group A each drank 800 mL of distilled water, while those in group B each drank 800 mL of isotonic saline. Their urine output was then measured at 60-minute intervals. The average results are recorded in the table below.

Time (min)	Urine output (mL min^{-1})	
	Group A	Group B
0	1.0	0.9
60 (drink taken)	0.9	1.0
120	12.4	1.5
180	4.3	1.9
240	2.1	2.1
300	1.1	1.7

(a) Describe the physiological responses that led to the changes in the urine output of group A participants. (5 marks)

© 2019 Aristo Educational Press Ltd.

(b) Account for the results of group B participants. (3 marks)

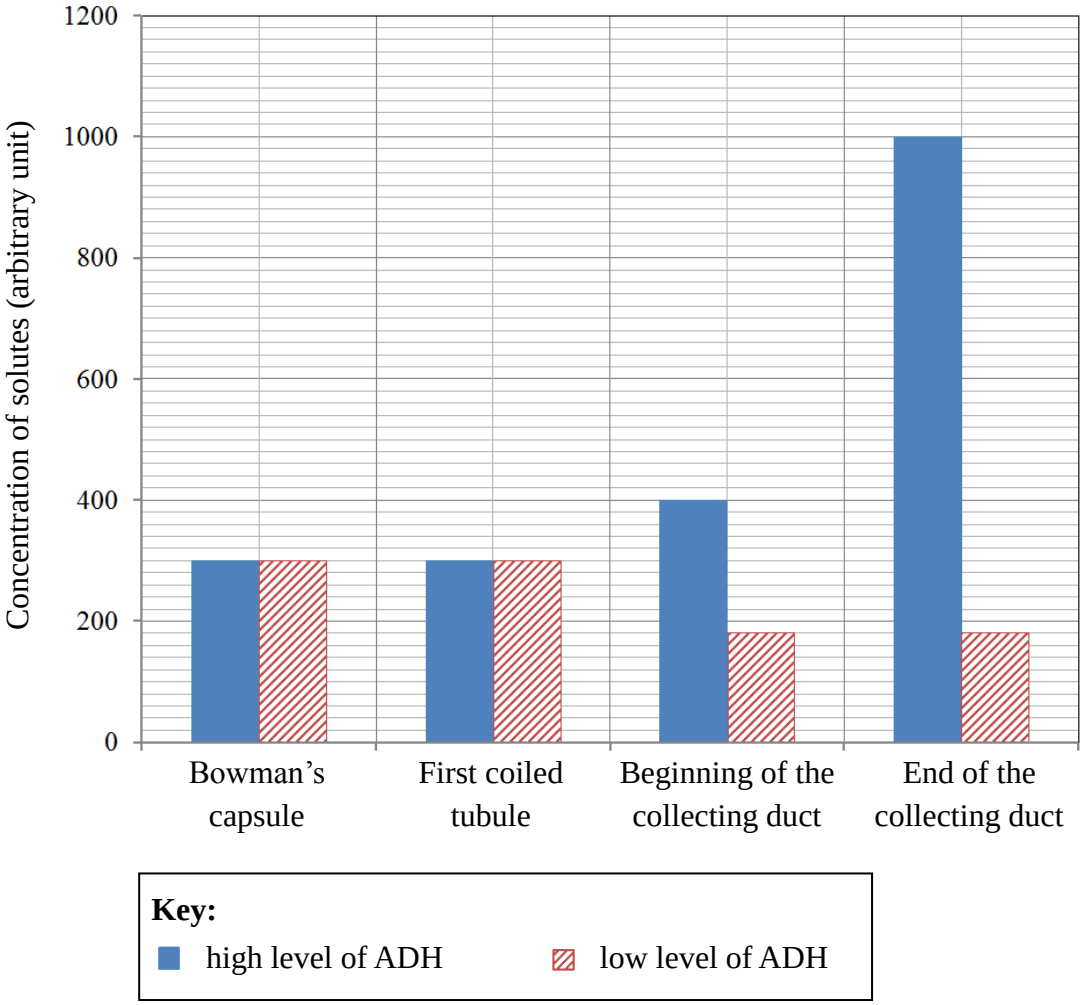
(c) If group A participants were injected with ADH 30 minutes after drinking distilled water, how would their urine output change? (3 marks)

(Total: 11 marks)

© 2019 Aristo Educational Press Ltd.

C. Exam challenge (9 marks)

★★ 13. The graph below shows the concentration of solutes in the glomerular filtrate in different regions of a nephron when the blood ADH level is high or low, respectively.



(a) Calculate the percentage change in the concentration of solutes between the Bowman's capsule and the end of the collecting duct when there is a high level of ADH in the blood. Show your working. (2 marks)

© 2019 Aristo Educational Press Ltd.

(b) Compare the changes in the concentration of solutes that occur when the level of ADH in the blood is high with the changes that occur when the level of ADH is low. (4 marks)

(c) With reference to the graph, explain how a rise in the level of ADH results in the formation of concentrated urine. (3 marks)

(Total: 9 marks)

© 2019 Aristo Educational Press Ltd.

Marking scheme

A. Concept checking (10 marks)

1.	B	2.	C	3.	C	4.	B	5.	C
6.	B	7.	C	8.	C	9.	D	10.	B

(1 mark each)

B. Problem solving (21 marks)

11. (a) (i) Active transport (1)
All glucose molecules present in the glomerular filtrate are reabsorbed. (1)
(ii) Diffusion (1)
Half of the urea molecules present in the glomerular filtrate is reabsorbed. (1)

(b) The size of proteins is too large to pass through the walls of the glomerulus and the Bowman's capsule. (1)

(c) As water in the filtrate is absorbed back into the blood, the amount of water left in the filtrate decreases. (1)
As the filtrate flows along the nephron, the friction with the walls of the tubule reduces the flow rate. (1)

(d) The rate of respiration is lower at 30 ^{o}C than at 37 ^{o}C, so the production of energy (ATP) decreases. (1)
The rate of active transport of glucose decreases as less energy is available. (1)
Therefore, the rate of reabsorption is not fast enough to reabsorb all the glucose from the filtrate. (1)

(Total: 10 marks)

12. (a) The ingested distilled water in the intestines is absorbed into the blood. The water potential of the blood increases. (1)
The rise in water potential is detected by osmoreceptors in the hypothalamus. (1)
The hypothalamus signals the pituitary gland to release less ADH. (1)
The walls of the collecting ducts become less permeable to water. (1)
A smaller proportion of water in the glomerular filtrate is reabsorbed into the blood. (1)
As a result, a larger volume of urine is formed.

(b) The ingestion of isotonic saline leads to an increase in blood volume but does not affect the water potential of the blood. (1)
The increased blood volume causes an increased hydrostatic pressure in the glomerulus. (1)
The rate of glomerular filtrate formation increases and hence the urine output increases. (1)

© 2019 Aristo Educational Press Ltd.

(c) ADH increases the permeability of the walls of the collecting ducts to water. (1)
A large proportion of water in the glomerular filtrate would be reabsorbed into the blood. (1)
The urine output would become more or less the same as the normal level. (1)

(Total: 11 marks)

C. Exam challenge (9 marks)

13. (a) (1000−300) ÷ 300 × 100% (1)
= 233% increase (1)

(b) An overall increase in concentration under high ADH level, while a decrease in concentration under low ADH level (1)
Between the Bowman's capsule and first coiled tubule, the concentrations of solutes remained the same under both conditions. (1)
Between the first coiled tubule and the beginning of the collecting duct, the extent of decrease in solute concentration under low ADH level is greater than the extent of increase under high ADH level. (1)
Between the beginning and the end of the collecting duct, the solute concentration rises under high ADH level, but it decreases / stays the same under low ADH level. (1)

(c) The solute concentration of fluid increases significantly along the collecting duct. (1)
A high ADH level increases the permeability of collecting ducts to water. (1)
A large proportion of water is reabsorbed by osmosis. (1)

(Total: 9 marks)

© 2019 Aristo Educational Press Ltd.